



RV College of Engineering®
(An autonomous institution affiliated to VTU, Belagavi)

Department of Master of Computer Applications

Date: 16.03.2026

IV SEMESTER – MCA492L Technical Seminar
Metaverse – Technology and Future Scope

The concept of the Metaverse will refer to a persistent, immersive virtual environment where users will interact with each other and digital objects using avatars in real time. It will be built using advanced technologies such as Virtual Reality (VR), Augmented Reality (AR), Artificial Intelligence (AI), and Blockchain. The Metaverse will be applied across various domains including gaming, education, healthcare, virtual workplaces, and e-commerce, making it a significant component of modern digital transformation. With increasing investments from major technology companies and growing interest in Web 3.0 technologies, the Metaverse will emerge as a key trend in the future of computing.

The seminar will review existing technologies and frameworks that contribute to the development of the Metaverse, including 3D virtual environments, blockchain-based digital ownership systems, and AI-driven interactive systems. Key concepts such as decentralized platforms, Non-Fungible Tokens (NFTs), and immersive user interfaces will be examined. Several platforms like virtual gaming environments and social virtual spaces have already demonstrated early implementations of Metaverse concepts. Existing research highlights advantages such as enhanced user engagement and new economic models, while also identifying limitations including high hardware costs, privacy concerns, scalability challenges, and lack of standardization.

The seminar will focus on analyzing the technological foundations, applications, and future scope of the Metaverse through a comprehensive literature survey and comparative study of existing systems. It will examine how core technologies like VR, AR, AI, and Blockchain contribute to building immersive environments. The study will aim to provide insights into the benefits, challenges, and real-world use cases of the Metaverse. The expected outcome will include a clear understanding of current developments, identification of limitations in existing approaches, and exploration of future research directions such as improved security, scalability, and user experience. The seminar will contribute to developing a strong conceptual understanding and critical analysis of the Metaverse as an emerging computing paradigm.

Signatures

Student

Internal Guide

**Committee Member/
External Evaluator**

Director